

EHRHART THEORY OF ZIGZAG (FENCE) POSET ORDER POLYTOPES

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ABSTRACT. We study the Ehrhart polynomials and h^* -vectors of order polytopes of fence (zigzag) posets. Using the fast computation methods from [?], we tabulate h^* -vectors for $n \leq 12$ and discover that the Ehrhart evaluations at fixed argument s satisfy linear recurrences whose characteristic polynomials have a closed-form generating function of Chebyshev type. We connect these polytopes to flagged Schur functions via embeddings into Gelfand–Tsetlin patterns.

1. INTRODUCTION

Let F_n denote the *fence (zigzag) poset* on n elements, with cover relations $1 <_{F_n} 2 >_{F_n} 3 <_{F_n} 4 >_{F_n} \cdots$. The *order polytope* $\mathcal{O}(F_n)$ [?] consists of all weakly order-preserving maps $F_n \rightarrow [0, 1]$.

These polytopes are Gorenstein (the fence is self-dual), so their h^* -vectors are palindromic. Our computations show they are also ultra-log-concave and real-rooted for all $n \leq 12$.

Remark 1.1. See also the recent work of Brändén and Solus [?] on h^* -vectors of zigzag order polytopes, which establishes real-rootedness and connections to Chebyshev polynomials.

2. h^* -VECTORS

Table ?? lists the h^* -vectors of $\mathcal{O}(F_n)$ for $n = 1, \dots, 12$, computed via the frontier DP with Ehrhart–Macdonald reciprocity from [?]. All are palindromic (the fence poset is self-dual, so the order polytope is Gorenstein) and have positive, ultra-log-concave coefficients. The h^* -polynomial has degree $n - 2$ for all $n \geq 2$.

3. A RECURRENCE FOR EHRHART EVALUATIONS

Let $L_n(s) = L_{\mathcal{O}(F_n)}(s)$ denote the Ehrhart polynomial of $\mathcal{O}(F_n)$ evaluated at a non-negative integer s . Computations up to $n = 25$ reveal the following recurrence.

Observation 3.1. *For each fixed $s \geq 2$, the subsequences $(L_{2m+1}(s))_{m \geq 0}$ and $(L_{2m}(s))_{m \geq 1}$ each satisfy a linear recurrence with constant coefficients given*

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TABLE 1. h^* -vectors of order polytopes of fence posets.

n	$h^*(t)$
1	1
2	1
3	$1 + t$
4	$1 + 3t + t^2$
5	$1 + 7t + 7t^2 + t^3$
6	$1 + 14t + 31t^2 + 14t^3 + t^4$
7	$1 + 26t + 109t^2 + 109t^3 + 26t^4 + t^5$
8	$1 + 46t + 334t^2 + 623t^3 + 334t^4 + 46t^5 + t^6$
9	$1 + 79t + 937t^2 + 2951t^3 + 2951t^4 + 937t^5 + 79t^6 + t^7$
10	$1 + 133t + 2475t^2 + 12331t^3 + 20641t^4 + 12331t^5 + 2475t^6 + 133t^7 + t^8$
11	$1 + 221t + 6267t^2 + 47191t^3 + 123216t^4 + 123216t^5$ $+ 47191t^6 + 6267t^7 + 221t^8 + t^9$
12	$1 + 364t + 15393t^2 + 169416t^3 + 656683t^4 + 1019051t^5$ $+ 656683t^6 + 169416t^7 + 15393t^8 + 364t^9 + t^{10}$

by the nonzero entries of row $2s$ of the OEIS triangle A109466, $T(n, k) = (-1)^{n-k} \binom{k}{n-k}$.

For example:

- $s = 2$: row 4 gives $(1, -3, 1)$, so $L_{n+4}(2) = 3L_{n+2}(2) - L_n(2)$ for n of fixed parity. The subsequences $(2, 5, 13, 34, 89, \dots)$ and $(3, 8, 21, 55, 144, \dots)$ are bisections of the Fibonacci sequence.
- $s = 3$: row 6 gives $(-1, 6, -5, 1)$, so $L_{n+6}(3) = 5L_{n+4}(3) - 6L_{n+2}(3) + L_n(3)$.
- $s = 4$: row 8 gives $(1, -10, 15, -7, 1)$, yielding an order-4 recurrence (in steps of 2).

In general, the recurrence for fixed s has order s and its characteristic polynomial $p_s(x)$ is read from row $2s$ of the triangle. The interleaved sequence $(L_n(s))_{n \geq 1}$ does not satisfy a recurrence from this triangle.

The generating function for the characteristic polynomials is

$$\sum_{s \geq 0} p_s(x) y^s = \frac{1 + y}{1 + (2 - x)y + y^2}, \quad (1)$$

or equivalently, the p_s satisfy the three-term recurrence

$$p_s(x) = (x - 2)p_{s-1}(x) - p_{s-2}(x), \quad p_0 = 1, \quad p_1 = x - 1.$$

4. CONNECTION TO FLAGGED SCHUR FUNCTIONS

Remark 4.1. The order polytope of the fence poset on n elements can be realized as a face of a Gelfand–Tsetlin polytope: the zigzag path in the fence embeds as a path through consecutive rows of a GT pattern of rectangular shape, with all entries not on the path frozen by the flag conditions. This

connects the fence Ehrhart polynomial to flagged Schur functions and may explain the clean Chebyshev-type structure of (??).

REFERENCES

- [1] P. Alexandersson, *Fast computation of Ehrhart polynomials of Gelfand–Tsetlin polytopes via Macdonald reciprocity*, preprint, 2025.
- [2] P. Brändén and L. Solus, *Symmetric decompositions, triangulations and real-rootedness*, preprint, 2025. <https://arxiv.org/abs/2503.16403>
- [3] R. P. Stanley, *Two poset polytopes*, Discrete Comput. Geom. **1** (1986), 9–23.

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